

HIKET — Bayesian Calibration of Forest SOC Models

Methods and pipeline documentation

Lorenzo Menichetti

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1 Overview

1.1 Research question

Finnish forest soils show a positive trend in mineral SOC stocks from the first systematic measurement campaign (VMI8, 1985–86) through Biosoil (2006) and into the Komeetta campaign (2024). The trend is modest but consistent, and its attribution is politically consequential: Finland reports it as a genuine sink, whereas comparable northern countries do not.

Existing SOC decomposition models, when run with published parameter sets or calibrated with standard steady-state assumptions, tend to predict SOC saturation or decline that is difficult to reconcile with the observed accumulation. This raises the central question of HIKET:

Do divergent model projections of Finnish forest SOC — including apparent saturation — reflect genuine structural differences between models, or artefacts of how the models are calibrated?

HIKET addresses this by calibrating six models against the same Finnish data under identical likelihood, error, filtering, and — crucially — initialisation assumptions. Residual differences among models are then attributable to structure rather than to parametric choices. A key innovation is the *transient initialisation* approach (§8), which gives all models a physically grounded, calibrated initial state in 1985, rather than the analytically-imposed steady state traditionally assumed. This single change substantially improves each model’s ability to track the observed SOC trajectory.

1.2 Models

Six models span a complexity ladder crossed with two climate-integration choices:

Model	Pools	Free params	Climate ξ	Notes
SP1	1	6	single triplet, Yasso07 form	Pure R
TP2	2	8	single triplet, Yasso07 form	ICBM topology, pure R
TP3	3	10	single triplet, Yasso07 form	ASH cascade, pure R
Yasso07	5	20	single triplet	Fortran core
Yasso15	5	26	pool-group-specific	Fortran core (yasso15.so)
Yasso20	5	26	pool-group-specific, monthly T	Same Fortran as Yasso15

Two comparison axes run through this design. First, a *pool complexity* axis: SP1 → TP2 → TP3 → Yasso07 increases the number of pools from 1 to 5 while maintaining the same Yasso07 climate response, isolating the effect of structural complexity on the calibrated SOC trajectory. Second, a *climate-integration* axis: Yasso07 → Yasso15 → Yasso20 adds pool-group-specific climate responses and finer intra-annual temperature integration, isolating the effect of climate-coupling refinements.

All models share the same AWEN litter fractionation, observation error model, likelihood form, and transient initialisation protocol. Residual differences in calibrated trajectories are therefore structural.

1.3 Pipeline

The pipeline runs in four sequential stages. Data preparation is documented separately in `Data_work.R`.

Script	Stage	Purpose
<code>run_{Model}_baseline.R</code>	0 — Baseline	Forward run at published defaults; no calibration
<code>run_{Model}_transient_calibration.R</code>	1 — Calibration	MCMC posterior sampling
<code>run_{Model}_transient_predictive.R</code>	2 — Predictive	Posterior predictive trajectories, residuals
<code>run_residual_analysis.R</code>	3 — Residuals	Random-forest residual analysis
<code>run_multimodel_comparison.R</code>	4 — Comparison	Cross-model metrics and overlay figures

Shared MCMC settings are defined in `calib_config.R`. Prior specifications live in `Prior_specs/{Model}_priors.R` sourced at the top of each calibration script.

2 Data

A brief data description is given here; the complete treatment is documented in `Data_work.R`.

2.1 Plots and SOC observations

The calibration dataset covers approximately 350–400 permanent Finnish NFI plots classified as calibration-ready (`calib_ready = TRUE` in `site_raw.csv`). Plots are excluded if they have zero or missing litter inputs, absent organic-layer measurements (OFH), or implied mean residence times exceeding 100 years.

Three SOC measurement campaigns provide the calibration targets:

Campaign	Year(s)	Notes
VMI8	1985–86	First national campaign; mineral SOC depth
Biosoil	2006	European network; same depth protocol
Komeetta	2024	Third Finnish campaign; confirms saturation

With three temporally separated observations per plot, the calibration constrains both the 1985–2006 accumulation phase and the subsequent 2006–2024 near-stability. This separation is important: models that happen to fit the first interval but diverge over the second are penalised.

2.2 Litter inputs

Annual AWEN-fractionated litter inputs ($\text{tC ha}^{-1} \text{ yr}^{-1}$) come from the plot-level dataset of Tupek et al. (Zenodo, DOI: 10.5281/zenodo.19736499). These are plot-level estimates and are approximately flat over the observation period — a key empirical constraint discussed in §8.

2.3 Climate

Gridded daily climate data (temperature, precipitation) span 1961–2025 (`nfi_plot_weather_data_1961_2025.nc`). Annual or monthly summaries are computed per plot according to each model’s climate-integration requirements.

3 Models

All six models follow first-order kinetics: each pool C_i decomposes at a rate proportional to its size, modulated by a dimensionless climate modifier $\xi \geq 0$. Litter input J is AWEN-fractionated (acid-, water-, ethanol-, non-soluble) and the fractionation is fixed from the input data.

3.1 SP1 (single pool)

All litter inputs — NWL, FWL, and CWL summed across AWEN fractions — enter a single pool C :

$$\frac{dC}{dt} = J(t) - \alpha \xi(t) C(t).$$

The annual step has a closed-form solution (piecewise-constant J and ξ within each year):

$$C(t+1) = C_{ss}(t) + (C(t) - C_{ss}(t)) e^{-\alpha \xi(t)}, \quad C_{ss}(t) = \frac{J(t)}{\alpha \xi(t)}.$$

Pure R; no Fortran.

3.2 TP2 (two pools)

Two pools, A (active) and H (humus). All litter enters A ; H receives transferred mass from A only:

$$\frac{dA}{dt} = J(t) - \alpha_A \xi(t) A(t), \quad \frac{dH}{dt} = p_H \alpha_A \xi(t) A(t) - \alpha_H \xi(t) H(t).$$

This is the ICBM topology (Andr n & K tterer 1997) with ξ multiplying both pool decay rates. Parameter naming follows Yasso07, making the TP2 parameter set a strict subset of Yasso07. The annual step is solved exactly (see  3.1), not Euler-approximated. Pure R.

3.3 TP3 (three pools)

Three pools in a strict sequential cascade: Active (A) $\rightarrow$ Slow (S) $\rightarrow$ Humus (H). All litter enters A ; a fraction p_S of A decomposition transfers to S ; a fraction p_H of S decomposition transfers to H . H has no climate modifier (consistent with TP2 and Yasso conventions). The discrete annual dynamics are:

$$\begin{aligned} A(t+1) &= A(t)(1 - \alpha_A \xi(t)) + J(t) \\ S(t+1) &= S(t)(1 - \alpha_S \xi(t)) + p_S \alpha_A \xi(t) A(t) \end{aligned}$$

$$H(t+1) = H(t)(1 - \alpha_H) + p_H \alpha_S \xi(t) S(t).$$

The Euler stepping is intentional: it matches the stepping used in the runtime simulation, so the initialisation pre-run and the forward run are numerically consistent. Analytical steady states exist and are used as the starting point for the transient pre-run:

$$A^* = \frac{J}{\alpha_A \xi}, \quad S^* = \frac{p_S J}{\alpha_S \xi}, \quad H^* = \frac{p_H p_S J}{\alpha_H}.$$

TP3 sits between TP2 and Yasso07 on the complexity ladder, adding a slow intermediate pool but retaining a sequential (no back-transfer) topology. Pure R.

3.4 The Yasso model family: shared structure

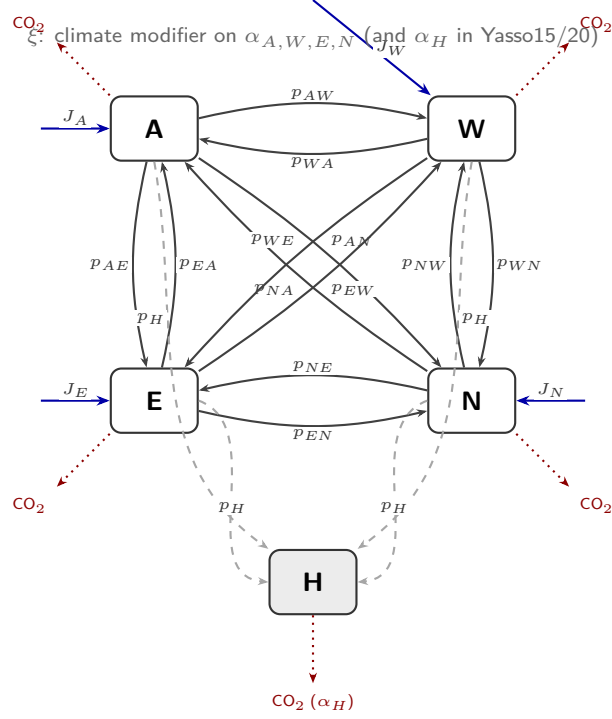
Yasso07, Yasso15, and Yasso20 share the same five-pool AWEN+H topology and the same Fortran matrix-exponential solver. They differ only in how the climate modifier ξ is computed (§3.5–3.7). This subsection describes the shared architecture for reference.

Pool structure and notation

The five pools are: **A** (acid-hydrolysable), **W** (water-soluble), **E** (ethanol-soluble), **N** (non-soluble, largely lignin), and **H** (humus). Litter enters the four AWEN pools in proportions fixed by the chemical fractionation of the input data. Carbon flows between AWEN pools at rates $p_{ij} \cdot \alpha_j \cdot \xi \cdot C_j$, where p_{ij} is the fraction of pool j 's decomposition flux routed to pool i . A fixed fraction p_H of each AWEN pool's decomposition flux forms humus; the remainder is respired as CO_2 . Humus decomposes irreversibly at rate α_H (no climate modifier in Yasso07; pool-group-specific ξ_H in Yasso15/20).

Pool diagram

The diagram below shows all pools, litter inputs, inter-pool transfers (labelled with HIKET parameter names), humification, and respiration pathways. Litter inputs J_A, J_W, J_E, J_N are AWEN fractions of the annual plot-level litter; for FWL and CWL cohorts a woody-size modifier $h_S(d)$ additionally scales α_j .



Arrow key: $\longrightarrow$ ~inter-pool transfer (p_{ij}) $\dashrightarrow$ ~humification (p_H) $\cdots\rightarrow$ ~respiration (CO_2)
 $\longrightarrow$ ~litter input (J)

Parameter summary

Parameter	Symbol	Role
Decomp. rates (AWEN)	$\alpha_A, \alpha_W, \alpha_E, \alpha_N$	Base first-order rate of each AWEN pool (yr^{-1}); fixed at published
Decomp. rate (H)	α_H	Humus decomposition rate (yr^{-1}); fixed
Transfer fractions	$p_{ij}, i \neq j \in \{A, W, E, N\}$	Fraction of pool j decomp. flux routed to pool i ; 12 per model, all fi
Humification	p_H	Fraction of each AWEN decomp. flux routed to H; same for all four
AWE climate, linear	β_1	Linear temperature coefficient in ξ_{AWE} ; free, log-transformed
AWE climate, quadratic	β_2	Quadratic temperature coefficient; free, unconstrained
AWE precipitation	γ	Precipitation saturation exponent in ξ_{AWE} ; free, unconstrained
N-pool climate	$\beta_{N1}, \beta_{N2}, \gamma_N$	As above, for the N pool separately
H-pool climate	$\beta_{H1}, \beta_{H2}, \gamma_H$	As above, for the H pool separately
Size modifier	δ_1, δ_2, r	Reduce α for woody cohorts: $h_S(d) = \min(1, (1 + \delta_1 d + \delta_2 d^2)^{- r })$; fi
Leaching	$w_1, \dots, w_5$	Leaching loss weights; inactive (leac=0) throughout; fixed
Historical litter	σ_{init}	Scales 1917 litter anchor in the pre-run; encodes pre-1985 productiv
Contemporary litter	σ_{input}	Scales observed litter inputs (1985 onward); absorbs litter model un

{Y07 = Yasso07; Y15 = Yasso15; Y20 = Yasso20; Y15/20 = both.}

3.5 Yasso07

Five pools (Tuomi et al. 2009, 2011): A (acid-soluble), W (water-soluble), E (ethanol-soluble), N (non-soluble / lignin), H (humus). Litter inputs enter A, W, E, N proportionally to AWEN fractions; transfers between AWEN pools are controlled by 12 fractions p_{ij} ; all four AWEN pools contribute to H at a humification rate p_H ; H decomposes irreversibly at rate α_H .

The dynamics are $dC/dt = AC + b(t)$, with the matrix A assembled from decomposition rates and transfer fractions. The Fortran core solves each annual step analytically via the matrix exponential $C(t+1) = A^{-1}(e^{A\Delta t}(AC(t)+b)-b)$,

with a Euler fallback for near-singular matrices. NWL, FWL, and CWL litter types are tracked as three parallel cohorts; the per-cohort state vector has 15 elements (5 pools $\times$ 3 cohorts).

The climate modifier uses a four-point Gaussian approximation of the intra-annual temperature cycle:

$$\xi(\bar{T}, T_{\text{amp}}, P) = \frac{1}{4} \sum_{k=1}^4 \exp(\beta_1 T_k + \beta_2 T_k^2) \times (1 - \exp(\gamma P/1000)),$$

where T_k are four quarterly temperatures derived from annual mean $\bar{T}$ and seasonal amplitude T_{amp} . A woody-size modifier scales decomposition for FWL/CWL cohorts:

$$h_S(d) = \min\left\{1, (1 + \delta_1 d + \delta_2 d^2)^{-|r|}\right\},$$

where d is the cohort diameter (cm; fixed at 2 cm for FWL, 15 cm for CWL).

3.6 Yasso15

Identical pool topology, AWEN partitioning, and matrix-exponential machinery as Yasso07. The structural change is *pool-group-specific* climate response: three triplets $(\beta_1, \beta_2, \gamma)$, $(\beta_{N1}, \beta_{N2}, \gamma_N)$, $(\beta_{H1}, \beta_{H2}, \gamma_H)$ act on the AWE, N, and H pool groups respectively. The functional form matches Yasso07 (four-point Gaussian temperature approximation, exponential-with-saturation precipitation response). Leaching parameters $w_1, \dots, w_5$ are present in the 35-element parameter vector but are inactive throughout (leaching scalar held at zero).

3.7 Yasso20

Same Fortran core as Yasso15 (compiled as `yasso15.so`, reused). The single structural change is the intra-annual temperature integration: instead of the four-point Gaussian approximation, the modifier is averaged directly over all 12 monthly temperatures:

$$\xi_T^{(g)} = \frac{1}{12} \sum_{m=1}^{12} \exp(\beta_1^{(g)} T_m + \beta_2^{(g)} T_m^2), \quad g \in \{\text{AWE}, N, H\}.$$

This is the only computational difference from Yasso15. Monthly climate data (12 rows per year per plot) are required, compared to annual summaries for all other models.

Yasso20 inherits the same 35-parameter structure as Yasso15 in this calibration (all 12 inter-pool flow fractions free; see §5.7).

Note on the published Yasso20 parameterisation. The MAP parameter values from the original global calibration (Viskari et al. 2022) include six transfer fractions constrained to zero. When these published values are applied directly, the per-pool mass budget for pools A and N is marginally exceeded (transfer fractions plus humification fraction sum slightly above 1.0), producing a numerically unstable forward simulation in the HIKET Fortran implementation, which enforces strict diagonal dominance. The HIKET calibration avoids this region of parameter space entirely through the stick-breaking budget constraint (§5.8). The Yasso20 uncalibrated baseline uses FMI prior-centre values, which satisfy mass conservation.

4 Fortran implementation: differences from the FMI reference code

The HIKET Fortran cores (`yasso07.f90`, `yasso15.f90`) derive from the FMI Yasso implementations (Järvenpää 2015 for Yasso15/20; Tuomi et al. for Yasso07) but incorporate several modifications introduced in 2025. These are documented here for transparency and to assist cross-comparison with the original FMI code.

4.1 Diagonal dominance check and Euler fallback

The annual timestep in each model is solved via the matrix exponential $\exp(A\Delta t)$. The original FMI code applies the matrix exponential unconditionally. The HIKET implementation adds a pre-check for diagonal dominance: for each column j of A , if the sum of off-diagonal absolute values (outflows to other pools and to humus) exceeds 99.99% of the diagonal absolute value (total decomposition rate), the matrix is classified as near-singular and a first-order Euler step is used instead:

$$C(t + \Delta t) \approx \max(0, C(t) + (A C(t) + b) \Delta t).$$

The condition being detected is mathematically equivalent to the per-pool mass budget $\sum_i p_{ij} + p_H \gtrsim 1$: the sum of all transfer fractions and the humification fraction approaches or exceeds 1, leaving zero or negative respiration from pool j . The $\max(0, \cdot)$ clamp prevents negative pool values in the Euler path but does not conserve mass exactly; repeated Euler steps with clamping can cause slow carbon drift (see §3.7 for the Yasso20 context). The HIKET calibration avoids this regime entirely via the stick-breaking budget constraint.

4.2 Matrix exponential: scaling cap and Taylor terms

The matrix exponential is computed by scaling-and-squaring with a Taylor series expansion. Two changes were made to the original implementation:

- **Scaling-step cap.** The original code had no upper bound on the number of scaling steps j (chosen so that $\|A\| < 2^j$). For ill-conditioned matrices with very large norms, j grew without bound and the squaring loop ran for thousands of iterations, effectively hanging. The cap $j \leq 64$ (covering $\|A\| \leq 2^{64} \approx 1.8 \times 10^{19}$) terminates the scaling phase for any physically meaningful matrix.
- **Taylor terms.** The number of Taylor series terms was increased from 10 to 20. When heavy scaling is applied (large j), the scaled matrix $A/2^j$ is close to zero and many terms are needed for the expansion to accumulate the correction accurately.

4.3 Terminal per-cohort state output (C_final)

The Fortran run routines (`yasso07_run_r`, `yasso15_run_r`) return a new output argument `C_final(15)`: the terminal per-cohort pool state $[C_{\text{nw1}} \mid C_{\text{fw1}} \mid C_{\text{cw1}}]$ at the end of the last simulated year. This argument is absent from the original FMI code, which returns only the five summed pool totals.

`C_final` is required for two purposes in HIKET. First, the transient pre-run chains directly into the calibration period: the pre-run's terminal state becomes C_{init} for 1985 (§8). Without the full 15-element per-cohort breakdown, re-initialising the forward simulation would require an analytical steady-state approximation that introduces additional error. Second, the 60-year forward projection (predictive stage) chains from the end of the historical run in the same way. The `C_final` field is attached to the R-side output data frame via `attr(run_out, "C_final")`.

4.4 ξ precomputation in R

The original FMI code computes the climate modifier ξ internally within the Fortran timestep loop, from raw temperature and precipitation inputs. HIKET separates this computation: ξ is computed as an annual (or monthly-aggregated) array in R *before* the Fortran call, and passed to the Fortran as a pre-computed vector argument. This separation has two benefits. First, it allows the Jensen's inequality correction (§8) to be applied consistently and transparently at the R level. Second, it makes the ξ functional form easily modifiable without recompiling Fortran; the different forms for Yasso07 (four-point Gaussian), Yasso15 (same form, pool-group-specific), and Yasso20 (12-monthly average) are handled in dedicated R functions (`compute_xi_yasso07`, `compute_xi_yasso15`, `compute_xi_yasso20`).

4.5 Floating-point precision (Yasso15 and Yasso20)

In `yasso15_mod`, the precision kind is defined as `dp = SELECTED_REAL_KIND(6, 37)`, which evaluates to `REAL(4)` (single precision, \$ 7 significant digits) on standard hardware. Despite the `REAL(dp)` declarations throughout the source, all Fortran arguments for Yasso15 and Yasso20 are therefore 4-byte. The R wrappers use `as.single()` for all `.Fortran()` calls to Yasso15/20.

Yasso07 uses `dp = SELECTED_REAL_KIND(15, 307)` = genuine double precision. Its R wrapper correctly uses `as.double()`.

An earlier version of the Yasso15 wrapper incorrectly used `as.double()` for the Yasso15/20 calls. This caused segfaults in the Fortran (type mismatch in the `.Fortran()` interface) that were resolved by reverting to `as.single()`.

4.6 Parameter ordering

The FMI `.dat` posterior files use a different parameter ordering from HIKET. In the FMI files, leaching weights w_1 – w_5 occupy positions 17–21, followed by the climate parameters; in HIKET, climate parameters occupy positions 17–25 and leaching weights are at 31–35. The remapping is applied in `Priors_model_matching.R` via the index vector `ORIG_TO_LORENZO` and verified by cross-checking MAP values against named CSV files (`y15par.csv`, `y07par_gui.csv`).

4.7 Yasso07: sign convention for r

The original Yasso07 parameter files store r as a negative value (e.g., -0.307). The Fortran applies $-|r|$ in the size modifier exponent, so the sign is irrelevant at runtime. HIKET stores r as positive and applies a log transform in the calibration (which requires positivity). The `abs()` conversion is applied when reading reference values from `y07par_gui.csv` or `YASSO07_DEFAULT_PARAMS`.

5 Parameter specification

5.1 Free, fixed, and derived parameters

For each model, parameters split into:

- **Free:** sampled by MCMC. Includes the two auxiliary uncertainty parameters σ_{init} and σ_{input} in every model.
- **Fixed:** held at published or structurally motivated values. For the Yasso family, decomposition rates and humus parameters are fixed at published defaults, reducing free-parameter count and keeping the intercomparison clean.
- **Derived:** none in the current pipeline.

Transforms map constrained physical parameters to an unconstrained sampling space, with log-Jacobian contributions added to the prior density:

- **log:** positivity; $x = \log \theta$.
- **logit:** $(0, 1)$ bounds; $x = \log(\theta/(1 - \theta))$.
- **unconstrained:** identity; no constraint beyond prior.
- **stick_break:** K -fraction simplex constrained to $\text{sum} \leq B$; described in §5.8.

5.2 SP1 (6 free, 0 fixed)

Parameter	Meaning	Transform
α	Decomposition rate (yr^{-1})	log
β_1	Temperature response, linear	log
β_2	Temperature response, quadratic	unconstrained
γ	Precipitation response	unconstrained
σ_{init}	Historical litter scaling (1917 anchor)	log
σ_{input}	Contemporary litter scaling	log

5.3 TP2 (8 free, 0 fixed)

Parameter	Meaning	Transform
α_A	A-pool decomposition rate (yr ⁻¹)	log
α_H	H-pool decomposition rate (yr ⁻¹)	log
p_H	Humification fraction (A → H)	logit
β_1, β_2, γ	Climate response	log, unc., unc.
$\sigma_{\text{init}}, \sigma_{\text{input}}$	Litter scalings	log, log

5.4 TP3 (10 free, 0 fixed)

Parameter	Meaning	Transform
α_A	A-pool decomposition rate (yr ⁻¹)	log
α_S	S-pool decomposition rate (yr ⁻¹)	log
α_H	H-pool decomposition rate (yr ⁻¹)	log
p_S	Transfer fraction A → S	logit
p_H	Transfer fraction S → H	logit
β_1, β_2, γ	Climate response	log, unc., unc.
$\sigma_{\text{init}}, \sigma_{\text{input}}$	Litter scalings	log, log

No fixed parameters. Like SP1 and TP2, TP3 has no published global calibration, so all parameters are learned from the Finnish data.

5.5 Yasso07 (20 free, 6 fixed)

Fixed (6): $\alpha_A, \alpha_W, \alpha_E, \alpha_N$ (AWEN decomposition rates), p_H, α_H . Published Tuomi et al. defaults.

Free (18):

Parameter	Count	Transform
12 transfer fractions p_{ij} , 4 columns of 3	12	stick_break
β_1, β_2, γ	3	log, unc., unc.
δ_1, δ_2, r	3	unc., log, log
$\sigma_{\text{init}}, \sigma_{\text{input}}$	2	log, log

$\delta_2 > 0$ enforced (Fortran NaN for $\delta_2 < 0$ at $d > 0.4$ cm). r stored as positive in HIKET; reference defaults store it negative.

5.6 Yasso15 (26 free, 11 fixed)

Fixed (11): $\alpha_A, \alpha_W, \alpha_E, \alpha_N, p_H, \alpha_H, w_1, \dots, w_5$ (leaching weights; inactive at leac = 0). Published Yasso15 defaults.

Free (28):

Parameter	Count	Transform
12 transfer fractions p_{ij}	12	stick_break
AWE climate: β_1, β_2, γ	3	log, unc., unc.
N climate: $\beta_{N1}, \beta_{N2}, \gamma_N$	3	log, unc., unc.

Parameter	Count	Transform
H climate: $\beta_{H1}, \beta_{H2}, \gamma_H$	3	log, unc., unc.
Size modifier: δ_1, δ_2, r	3	unc., log, log
$\sigma_{\text{init}}, \sigma_{\text{input}}$	2	log, log

5.7 Yasso20 (26 free, 11 fixed)

Identical fixed/free structure to Yasso15. The same 35-parameter vector is used; only the ξ computation method differs at runtime.

Uniform 12-fraction structure across the Yasso family. Earlier Yasso20 calibrations (Viskari 2022) fixed six transfer fractions at zero (structural zeros from a global-dataset MCMC) and derived three algebraically. HIKET does not impose these constraints. All 12 inter-pool fractions are sampled in all three Yasso models, for two reasons. First, the Viskari zeros reflect a warm-site global calibration and are not general biological laws; Finnish boreal data should be free to inform pool-flow structure. Second, fewer free fractions in Yasso20 than in Yasso07 or Yasso15 would confound structural (climate-integration) differences with parametric (degrees of freedom) differences, making the intercomparison uninterpretable. Prior centres for fractions use the published Yasso15 defaults (see §4).

5.8 Stick-breaking parameterisation for transfer fractions

Each column j of the AWEN transfer matrix has three outflow fractions satisfying $\sum_i p_{ij} + p_H \leq 1$ (mass conservation; the unallocated remainder $1 - \sum_i p_{ij} - p_H$ is respired as CO_2). The constraint is enforced via stick-breaking with budget $B = 1 - p_H$. Three unconstrained parameters z_1, z_2, z_3 map to the three fractions by sequentially breaking pieces off the remaining budget:

$$p_1 = B \cdot \sigma(z_1), \quad p_2 = (B - p_1) \cdot \sigma(z_2), \quad p_3 = (B - p_1 - p_2) \cdot \sigma(z_3),$$

where σ is the logistic function. Because each $\sigma(z_i) < 1$, the sum $p_1 + p_2 + p_3$ is strictly less than B , guaranteeing a strictly positive respired fraction. All three fractions in a column are therefore free parameters — none is algebraically determined — and the log-Jacobian of the transform is added to the prior. This guarantees the budget constraint exactly and produces a well-conditioned Jacobian. Each of the four AWEN columns uses an independent stick-breaking transform with the same budget B , contributing $4 \times 3 = 12$ free parameters for the 12 transfer fractions.

5.9 Cross-model parameter overlap

Parameter	SP1	TP2	TP3	Yasso07	Yasso15	Yasso20
α (single rate)	free	—	—	—	—	—
α_A, α_H	—	free	free	fixed	fixed	fixed
α_S	—	—	free	—	—	—
$\alpha_W, \alpha_E, \alpha_N$	—	—	—	fixed	fixed	fixed
p_H	—	free	free	fixed	fixed	fixed
p_S	—	—	free	—	—	—
12 transfer fractions p_{ij}	—	—	—	free	free	free
β_1, β_2, γ	free	free	free	free	free (AWE)	free (AWE)
$\beta_{N1}, \beta_{N2}, \gamma_N$	—	—	—	—	free	free
$\beta_{H1}, \beta_{H2}, \gamma_H$	—	—	—	—	free	free
δ_1, δ_2, r	—	—	—	free	free	free
$w_1, \dots, w_5$	—	—	—	—	fixed	fixed
$\sigma_{\text{init}}, \sigma_{\text{input}}$	free	free	free	free	free	free
Total free	6	8	10	20	26	26

6 Priors

All priors are normal in unconstrained (transformed) space, centred at published model defaults. The joint prior factorises across parameters:

$$\log p(x) = \sum_i \log \mathcal{N}(x_i; x_i^{\text{ctr}}, \sigma_{\text{ppm},i}) + \log J(x),$$

where x^{ctr} is the unconstrained image of the physical-space prior centre, σ_{ppm} is the prior standard deviation in unconstrained space, and $\log J$ is the log-Jacobian of the transforms. Prior specifications are stored in `Prior_specs/{Model}_priors.R` and sourced at the top of each calibration script.

6.1 Sources

Model	Source for prior centres	Source for prior widths
SP1	Hand-set (no published global calib.)	Weakly informative
TP2	Hand-set ICBM-style defaults	Weakly informative
TP3	Hand-set; extends TP2 with Slow pool	Weakly informative
Yasso07	<code>y07par_gui.csv</code> (Tuomi et al.)	Literature CIs
Yasso15	<code>Yasso15.dat</code> posterior mean (FMI)	<code>Yasso15.dat</code> posterior SD
Yasso20	<code>Yasso20_sample_parameters.rda</code> (FMI)	<code>Yasso20.dat</code> posterior SD

The FMI repository is at <https://github.com/YASSOmodel/Ryassofortran/tree/master/data>. The `.dat` files contain 10,000 MCMC samples from the original calibrations (row 1 = MAP, rows 2–10,001 = posterior draws). Column ordering in the FMI files differs from HIKET’s parameter ordering; remapping is performed and verified in `Priors_model_matching.R`.

For Yasso07, no posterior file is publicly available; σ_{ppm} is derived from the confidence intervals reported in Tuomi et al. (2008–2010). For SP1, TP2, and TP3, priors are hand-set to plausible starting values and wide enough to let the Finnish data dominate.

6.2 Transfer-fraction priors

For all three Yasso models the prior standard deviation on every transfer fraction is $\sigma_{\text{ppm}} = 1.0$ in unconstrained (stick-breaking) space. This is deliberately weakly informative: pool-flow structure is the primary structural-comparison axis, and the Finnish data should drive it. Prior centres for the fractions use the published Yasso15 defaults for all three Yasso models (Yasso20 included, since the previously-fixed zeros are now free parameters and Yasso15 defaults are the closest available nonzero prior centres from the same chemical-fractionation framework).

6.3 Climate-response and size-modifier priors

Climate and size parameters receive *informative* priors, derived from the FMI posterior (Yasso15/20) or from literature CIs (Yasso07). This prevents two pathologies:

1. **The $\gamma, \sigma_{\text{input}}$ compensation loop.** Both parameters scale total decomposition; without an informative prior on γ they trade off freely, producing a ridge in the posterior that destabilises MCMC mixing.
2. **The $\delta_2 < 0$ NaN pathology.** The Fortran size modifier returns NaN for $\delta_2 < 0$ at $d > 0.4$ cm; a log transform with informative prior keeps δ_2 in the well-behaved region.

γ_H for Yasso15 is capped at $\sigma_{\text{ppm}} = 1.5$: the H-pool precipitation response is near-unidentifiable at Finnish precipitation levels (~ 600 mm). The large raw posterior SD (~ 3) reflects a flat likelihood surface, not genuine uncertainty.

6.4 Numerical values

Yasso07

Parameter	x^{ctr} (physical)	σ_{ppm}	Space
p_{ij} (12)	published defaults	1.0	unconstrained
β_1	0.0987	0.20	log
β_2	-0.00157	0.05	physical
γ	-1.272	0.30	physical
δ_1	-1.708	0.15	physical
δ_2	0.859	0.10	log
r	0.307	0.015	log
$\sigma_{\text{init}}, \sigma_{\text{input}}$	1.00	0.50	log

Yasso15

Parameter	x^{ctr}	σ_{ppm}	Space
p_{ij} (12)	published defaults	1.0	unconstrained
β_1	0.0906	0.0459	log
β_2	-2.15×10^{-4}	1.4×10^{-4}	physical
γ	-1.809	0.0702	physical
β_{N1}	0.0488	0.105	log
β_{N2}	-7.92×10^{-5}	8×10^{-5}	physical
γ_N	-1.173	0.155	physical
β_{H1}	0.0352	0.135	log
β_{H2}	-2.08×10^{-4}	1.6×10^{-4}	physical
γ_H	-12.54	1.50 (capped)	physical
δ_1	-0.439	0.328	physical
δ_2	1.267	0.286	log
r	0.257	0.055	log
$\sigma_{\text{init}}, \sigma_{\text{input}}$	1.00	0.50	log

Yasso20

Parameter	x^{ctr}	σ_{ppm}	Space
p_{ij} (12)	Yasso15 defaults	1.0	unconstrained
β_1	0.158	0.104	log
β_2	-0.002	5.4×10^{-4}	physical
γ	-1.440	0.146	physical
β_{N1}	0.170	0.0625	log
β_{N2}	-0.005	4.1×10^{-4}	physical
γ_N	-2.000	0.0353	physical
β_{H1}	0.067	0.0912	log
β_{H2}	0.000	9×10^{-5}	physical
γ_H	-6.900	1.391	physical
δ_1	-2.550	0.435	physical
δ_2	1.240	0.211	log
r	0.250	0.0545	log
$\sigma_{\text{init}}, \sigma_{\text{input}}$	1.00	0.50	log

SP1, TP2, TP3

Parameter	x^{ctr}	σ_{ppm}	Space
$SP1$			
α	0.09	0.50	log

Parameter	x^{ctr}	σ_{ppm}	Space
β_1, β_2, γ	0.095, -0.00014, -1.21	0.20, 0.05, 0.30	log, phys., phys.
<i>TP2</i>			
α_A, α_H	0.73, 0.0015	0.50, 0.50	log
p_H	0.028	1.00	logit
β_1, β_2, γ	0.095, -0.00014, -1.21	0.20, 0.05, 0.30	log, phys., phys.
<i>TP3</i>			
$\alpha_A, \alpha_S, \alpha_H$	0.73, 0.10, 0.0015	0.50, 0.50, 0.50	log
p_S, p_H	0.028, 0.50	1.00, 1.00	logit
β_1, β_2, γ	0.095, -0.00014, -1.21	0.20, 0.05, 0.30	log, phys., phys.
<i>All three</i>			
$\sigma_{\text{init}}, \sigma_{\text{input}}$	1.00	0.50	log

7 Uncalibrated baselines

Before examining what the Finnish data tells the models, it is useful to know what the models say *without* any Finnish information — that is, how well published global parameter sets predict Finnish SOC stocks. These uncalibrated baseline runs serve as both a diagnostic and a narrative anchor: they show where calibration starts and how far it moves the predictions.

7.1 Purpose and scope

Baseline scripts (`run_yasso07_baseline.R`, `run_yasso15_baseline.R`, `run_yasso20_baseline.R`) run each Yasso model forward at its published default parameters across all calibration-ready plots. No MCMC, no posterior — a single deterministic forward run per plot. Because SP1, TP2, and TP3 have no published global calibration, no baselines are produced for them; the natural reference for the simple models would be their prior centres, but such runs are not currently generated.

Each baseline uses a steady-state initial condition rather than the 68-year pre-run. At published defaults $\sigma_{\text{init}} = 1$ and $\sigma_{\text{input}} = 1$, so the steady state and the pre-run endpoint coincide; the simpler steady-state formulation is used for transparency. The two auxiliary uncertainty parameters are therefore not exercised in the baseline — their effect is, by construction, the identity.

7.2 Parameter choices

The baseline parameter vector for each model combines the published decomposition rates, transfer fractions, climate-response parameters, and size-modifier parameters from the source indicated below. Auxiliary parameters are fixed at $\sigma_{\text{init}} = \sigma_{\text{input}} = 1$.

Model	Transfer fractions	Climate / size parameters
Yasso07	Tuomi et al. published defaults (<code>y07par_gui.csv</code>)	Same source
Yasso15	Published Yasso15 defaults (<code>y15par.csv</code>)	Same source
Yasso20	Yasso15 published defaults	FMI MAP (<code>Yasso20_sample_parameters.rda</code>)

For Yasso07 and Yasso15 the baseline is the full published parameter set. For Yasso20, the climate and size parameters are the FMI MAP values, but the 12 transfer fractions use the Yasso15 published defaults rather than the Viskari (2022) Yasso20 fraction values. At the published Yasso20 fractions, the per-pool outflow for two pools marginally exceeds the available budget; the HIKET solver’s diagonal-dominance guard (see the Fortran-implementation section) classifies this as non-conservative and the run is numerically unstable. The Yasso15 fractions satisfy the budget and

run stably. This substitution applies only to the Yasso20 baseline; the Yasso20 calibration (§5.7) samples all 12 fractions freely under the budget constraint and is unaffected.

Status. The choice of Yasso20 baseline parameterisation is provisional. An alternative would be to reproduce the original Yasso20 baseline using the unmodified FMI reference code, which tolerates the published fractions without the diagonal-dominance guard. Which option to adopt is left open pending the manuscript narrative.

7.3 Diagnostic outputs

Each baseline produces the following outputs in `diagnostics/{Model}_baseline/`:

File	Content
<code>*_mean_soc_trajectory.png</code>	Mean predicted SOC $\pm$ 1 SE across plots vs observed campaign means
<code>*_delta_soc_trajectory.png</code>	Mean annual Δ SOC rate ($\text{tC ha}^{-1} \text{ yr}^{-1}$) vs observed
<code>*_obs_vs_pred.png</code>	Scatter coloured by vegetation type (<code>kasvup_tyyppi</code>)
<code>*_residuals_diagnostic.png</code>	6-panel residual diagnostics
<code>*_rf_importance.png</code>	Random-forest permutation importance of site covariates
<code>*_worst25_timeseries.png</code>	Time series for the 25 worst-fitting plots
<code>*_residuals_histogram.png</code>	Distribution of log-scale residuals
<code>*_residuals.csv</code>	Per-observation residuals
<code>*_plot_ranking.csv</code>	Per-plot mean absolute log-residual
<code>*_report.txt</code>	Summary metrics (R^2 , RMSE, bias, MAPE) and RF results

The mean absolute SOC and the annual Δ SOC trajectory plots use the same visual form as the calibrated predictive outputs, allowing direct side-by-side comparison of baseline and posterior trajectories in the manuscript figures.

8 Transient initialisation

8.1 Why initialisation matters for the Finnish data

The observed Finnish SOC trajectory — accumulating from 1985 through Biosoil (2006) and approximately stabilising toward Komeetta (2024) — is not easily reproduced by a model that starts from an analytically-imposed steady state in 1985. The reason is straightforward. Under first-order kinetics with approximately flat litter inputs (as observed in the plot-level Tupek et al. dataset), a model at steady state in 1985 predicts approximately flat SOC from that point forward. But the data shows accumulation. The only way a model at steady state can produce an upward SOC trend is if litter inputs are also rising — and the plot-level inputs do not show a rise of sufficient magnitude.

The implication is that Finnish forests in 1985 were *not* at equilibrium with their recent litter inputs. Their SOC stocks were below what the contemporary productivity would sustain at steady state, and they have been accumulating toward that higher equilibrium ever since. This is consistent with the management history of Finnish forests: the post-war decades saw intensive management (clearcutting, drainage, fertilisation) that left many forest stands in a recovering state by 1985, with carbon stocks still building up.

A calibration that forces $C(1985) = C_{ss}$ (contemporary litter) has no degree of freedom to fit this accumulation signal. Either it rejects all models, or it absorbs the discrepancy into implausible kinetic parameters. The transient initialisation approach removes this constraint.

8.2 The 68-year pre-run

Instead of computing $C(1985)$ analytically from contemporary inputs, each MCMC draw runs the model forward 68 years from 1917 to 1985, ending at a calibrated initial state.

The choice of 1917 (Finland’s independence) as the pre-run anchor is partly practical (a natural reference point for Finnish forestry history) and partly statistical (68 years is long enough that the fast AWEN pools are approximately at steady state by 1985 regardless of the start condition; only H retains meaningful memory of the initial state).

During the pre-run, two inputs must be specified:

Climate: No gridded climate data exist before 1961. The pre-run therefore uses a constant climate modifier equal to $\xi(\text{climate})$ — the modifier evaluated at the mean of the observed climate window (1985–2004). This is an approximation, but an appropriate one: the Yasso climate response is smooth and the mean is a reasonable representative value for this multi-decadal period.

Litter: Litter inputs are linearly interpolated between a historical anchor at 1917 and the start of the observed series at 1985:

$$J(i) = J_{1917} + \frac{i-1}{67} (J_{1985} - J_{1917}), \quad i = 1, \dots, 68.$$

The anchors are parameterised as:

$$J_{1917} = J_{\text{full mean}} \cdot \sigma_{\text{init}} \cdot \sigma_{\text{input}}, \quad J_{1985} = J_{t_0 \text{ mean}} \cdot \sigma_{\text{input}},$$

where $J_{\text{full mean}}$ is the mean litter input over the full observed series and $J_{t_0 \text{ mean}}$ is the mean over the first five observed years (1985–1989). The pre-run begins at the analytical steady state under J_{1917} and constant climate.

8.3 The two auxiliary uncertainty parameters

σ_{init} and σ_{input} are free parameters in every model, both log-transformed with a prior centred at 1.0 (log-normal, unconstrained-space SD = 0.5, physical 95% interval approximately [0.37, 2.72]).

σ_{input} scales the entire contemporary litter time series (1985 onward). It absorbs uncertainty in the litter input model — plot-level litter estimates carry uncertainty from both the litter model itself and the representation of understorey vegetation, which is not individually measured. σ_{input} also scales J_{1985} , the endpoint of the pre-run interpolation.

σ_{init} scales only the 1917 litter anchor, J_{1917} . It encodes the ratio of historical (pre-1985) to contemporary mean productivity. The two parameters have strictly non-overlapping roles in the pre-run formula:

$\sigma_{\text{init}} > 1$: forests were historically *more* productive (old-growth or pre-intensification conditions). The 1985 initial SOC is above what contemporary litter inputs would sustain at steady state. The model predicts declining SOC from 1985.

$\sigma_{\text{init}} < 1$: forests were historically *less* productive (post-disturbance recovery). The 1985 initial SOC is below the contemporary equilibrium. The model predicts accumulating SOC from 1985, consistent with the Finnish observations.

$\sigma_{\text{init}} = 1$: the 1917 litter was equal to the contemporary series mean; the pre-run approaches the same initial state as the analytic steady state.

The posterior on σ_{init} is therefore a direct inference about the pre-1985 productivity history of Finnish forest plots. A posterior that favours $\sigma_{\text{init}} < 1$ across models constitutes model-based evidence for the post-disturbance recovery narrative.

8.4 Jensen’s inequality correction

The steady-state climate modifier used to start the pre-run is computed as $\xi(\text{climate})$, not as $\overline{\xi(\text{climate}_t)}$. Because ξ is convex in temperature and concave in precipitation, Jensen’s inequality gives $\overline{\xi_t} \geq \xi(\text{climate})$; the bias is 5–15% under typical Finnish inter-annual variability. Each model wrapper provides a `compute_xi_mean_*` function that averages the climate scalars over the steady-state window (first 20 observed years) and evaluates ξ once at the means, correctly avoiding the average-of- ξ bias.

8.5 Implementation across models

The pre-run protocol is uniform but adapted to each model’s solver:

Model	Pre-run implementation
SP1, TP2	Analytical 68-step R loop (exact per-step solution)
TP3	Euler 68-step R loop (matches the runtime stepping method)
Yasso07/15/20	68-row litter input fed to the Fortran solver; terminal per-cohort state extracted via <code>attr(run, "C_final")</code>

For the Yasso Fortran models, the pre-run is implemented as a genuine Fortran run, not an analytical approximation. The 15-element terminal state (5 pools $\times$ 3 litter cohorts) is returned via `C_final`, an output argument added specifically to support this chaining. Without it, only the 5 summed pool totals would be available, which is insufficient to correctly initialise the per-cohort bookkeeping for the forward simulation.

8.6 Likelihood adjustment

In the non-transient version of the engine (retained as a fallback), σ_{init} inflated the observation standard deviation at the first observation per plot, acting as a likelihood patch to absorb initial-state uncertainty. With transient initialisation, σ_{init} has already entered the model dynamics; retaining the likelihood patch would double-count its effect. The engine flag `transient_init = TRUE` removes this patch.

9 Likelihood

9.1 Form

Conditional on parameters, observations are modelled as

$$y_{p,t} \sim \mathcal{N}(\hat{C}_{p,t}, \hat{C}_{p,t} \cdot \sigma_{\text{obs}}),$$

where $y_{p,t}$ is observed SOC (tC ha⁻¹) at plot p , year t , and $\hat{C}_{p,t}$ is predicted total SOC. This is a multiplicative-normal model: the error standard deviation scales with the prediction, so relative observation error is constant across the SOC range.

9.2 Fixed σ_{obs}

σ_{obs} is held at the empirical coefficient of variation of the SOC observations across all calibration plots and years (≈ 0.45 – 0.50). It is not sampled.

This is a deliberate design choice. If σ_{obs} were sampled, a structurally inadequate model could inflate it to absorb its residual variance, hiding structural inadequacy. By fixing σ_{obs} at the same empirical CV across all six models, structural differences are forced into the residuals — which is exactly where the intercomparison wants them.

9.3 Per-plot log-likelihood

For plot p with observations at $\{t_1, \dots, t_{n_p}\}$:

$$\log L_p = \sum_{k=1}^{n_p} \log \mathcal{N}(y_{p,t_k}; \hat{C}_{p,t_k}, \hat{C}_{p,t_k} \cdot \sigma_{\text{obs}}).$$

Total log-likelihood: $\sum_p \log L_p$ plus the log-Jacobian of parameter transforms. Plots are evaluated in parallel via `mclapply`.

9.4 Rejection logic

Three hard rejections (return $-\infty$) apply:

1. $\hat{C}_{p,t}$ is non-finite at any observation year.

2. $\hat{C}_{p,t} \leq 0$ at any observation year (undefined in the multiplicative-error model).
3. The model run returned `NULL` (Fortran crash or wrapper exception).

No physical-ceiling guard is applied. Implausible predictions yield large negative log-likelihoods and are rejected by MCMC naturally. The likelihood evaluates $\hat{C}_{p,t}$ only at observation years; trajectory values between campaigns are computed but not entered into the likelihood.

9.5 Holdout split for independent validation

Plots flagged `is_holdout` in `site_raw.csv` are excluded from the likelihood sum: the calibration is fit only to the non-holdout plots. The held-out plots are carried forward (via the inputs RDS) to the predictive stage, where prediction metrics (R^2 , RMSE, bias, 95% coverage) are computed separately for the calibration and holdout sets. This provides an out-of-sample check that the calibrated models generalise beyond the plots used to fit them, rather than only reproducing in-sample fit. The split is applied uniformly across all six models, using the same `is_holdout` plot assignment so that every model is validated against the same held-out plots.

10 MCMC sampling

The Differential Evolution with snooker update sampler (DEzs) is used, via the `BayesianTools` package (Hartig et al. 2023). DEzs proposes new states using weighted differences between current chain states, then accepts/rejects via the standard Metropolis ratio. It mixes well for moderate-dimensional problems (10–50 parameters) and handles correlated posteriors better than random-walk Metropolis.

10.1 Settings

Shared across all six models, defined in `calib_config.R`:

Setting	Value	Note
Chains	5	Sufficient for Gelman–Rubin $\hat{R}$
Iterations	50,000	Per chain
Burn-in	5,000	10% of iterations
Thinning	None	Full chain retained
Initial state	prior centre x^{ctr}	All chains seeded at same point

10.2 Parallelisation

Two levels of parallelism are available:

- **Per-plot, within-chain:** each likelihood evaluation runs the per-plot inner loop in parallel via `mclapply` with `CORES_PER_CHAIN` workers.
- **Across-chain:** chains can be launched in parallel on a multi-core machine, each running its own per-plot parallelism.

The within-chain mode is mandatory on macOS (nested forking is unsafe).

11 Sanity checks and diagnostics

11.1 Pre-MCMC sanity check

Before launching MCMC, each calibration script runs a forward simulation at the prior centre for four randomly selected plots. Three conditions must hold:

1. All predicted SOC values are finite and positive.
2. Predicted-to-observed ratio lies in $[0.1, 10]$ at every observation year.
3. The likelihood `ll_fn(best_x)` is finite across all calibration plots.

A diagnostic PNG is written (`{Model}_forward_at_defaults_{RUN\_ID}.png`). Condition 1 or 3 failure aborts the calibration; condition 2 produces a warning only.

Input diagnostics (`diagnostics_inputs.R`, run separately) confirm that implied mean residence times (MRT = mean SOC / mean litter) do not exceed 100 years. Plots exceeding this threshold are excluded upstream (`calib_ready = FALSE`).

11.2 Convergence diagnostics

Computed on post-burnin samples in unconstrained (sampling) space:

Diagnostic	Threshold	Interpretation
$\hat{R}$ (univariate)	< 1.05	Good per-parameter convergence
$\hat{R}$ (multivariate)	< 1.05	Good joint convergence
Effective sample size	> 200	Adequate independent draws
Acceptance rate	0.2–0.4	Reasonable DEzs mixing

With 20–26 free parameters in the Yasso models and only two to three SOC observations per plot, poor $\hat{R}$ for individual transfer fractions is expected — the fractions are weakly identified by the aggregate SOC likelihood. This is structural non-identifiability, not a sampler failure.

11.3 Diagnostic outputs

Written to `Calibration_real_data_transient/diagnostics/{Model}/:`

File	Content
<code>*_traces*.png</code>	Trace plots, one panel per parameter
<code>*_marginals*.png</code>	Prior vs posterior KDE (shared x-grid)
<code>*_rhat_barplot*.png</code>	$\hat{R}$ per parameter
<code>*_kl_divergence*.png</code>	KL divergence posterior $\parallel$ prior
<code>*_report*.txt</code>	Tabular $\hat{R}$, ESS, quantile summaries
<code>*_forward_at_defaults*.png</code>	Pre-MCMC sanity check (4 plots)

Prior and posterior KDE are evaluated on the same x-grid derived from the prior’s 99th-percentile range, so tight priors are not artificially compressed.

12 Output files

Written to `Calibration_real_data_transient/runs/:`

File	Content	Used by
<code>{Model}_chains_{RUN_ID}.rds</code>	BayesianTools chain objects (unconstrained space)	Re-diagnostics
<code>{Model}_posterior_{RUN_ID}.rds</code>	Post-burnin samples in physical space	Predictive scripts
<code>{Model}_metadata_{RUN_ID}.rds</code>	Run configuration, wallclock, $\hat{R}$, ESS	Reproducibility
<code>{Model}_inputs_{RUN_ID}.rds</code>	Per-plot input structures used in calibration	Predictive scripts

The inputs RDS contains `climate_by_plot`, `inputs_by_plot`, `litter_means`, `obs_meta`, `plot_info`, `plots_real`, `holdout_plots`, `sigma_obs_fixed`, `sigma_ppm`, `STEADY_STATE_YEARS`, and the transient pre-run parameters (`PREINIT_YEAR`, `N_PREINIT_SMOOTH`). It provides exact provenance: predictive scripts use the same per-plot structures the calibration used, regardless of any subsequent updates to `Data_work.R`.

13 Key design decisions

1. **Transient initialisation via a 68-year pre-run.** The 1985 SOC trajectory is inconsistent with a steady-state initial condition under flat litter inputs. The pre-run from 1917 gives all models a calibrated, physically-grounded initial state. σ_{init} is the key parameter encoding pre-1985 productivity history; its posterior is directly interpretable as evidence for or against a post-disturbance recovery narrative.
2. **Uniform pre-run protocol across all six models.** SP1/TP2 use exact analytical steps; TP3 uses Euler steps matching its runtime method; Yasso07/15/20 use the Fortran solver with `C_final` chaining. The protocol produces the same conceptual quantity (initial state at 1985) for all models; structural differences in the resulting posteriors are not contaminated by initialisation differences.
3. **Six models spanning two design axes.** Pool complexity (SP1 $\rightarrow$ TP2 $\rightarrow$ TP3 $\rightarrow$ Yasso07) and climate integration (Yasso07 $\rightarrow$ Yasso15 $\rightarrow$ Yasso20) vary independently. Conclusions about the saturation signal are drawn from the ensemble, not from individual models.
4. **Multiplicative-normal likelihood.** The standard deviation of the observation error scales with the prediction, representing proportional rather than absolute measurement uncertainty. This is appropriate for SOC stocks, where measurement noise and spatial representativeness both scale with stock magnitude.
5. **Fixed σ_{obs}** at the empirical CV of the SOC observations, identical across all six models. Prevents structurally inadequate models from absorbing their residual variance into the error term; forces differences into the residuals where the intercomparison can examine them.
6. **$\xi(\text{climate})$ rather than $\overline{\xi(\text{climate}_t)}$** for the pre-run climate. Jensen’s inequality correction; 5–15% bias in boreal conditions otherwise. Implemented uniformly via `compute_xi_mean_*` functions.
7. **Steady-state climate window = first 20 observed years.** The 1985–2004 climate is used to compute the mean ξ for the pre-run. This pre-dates the accelerated boreal warming of the 2010s and is representative of the pre-run period.
8. **Decomposition rates fixed at published defaults** for the Yasso family ($\alpha_A, \alpha_W, \alpha_E, \alpha_N, p_H, \alpha_H$). Reduces free-parameter count and keeps the intercomparison clean. SP1, TP2, and TP3 have no published global calibrations; all their parameters are free.
9. **All 12 transfer fractions free in all three Yasso models.** The Viskari (2022) structural zeros and derived fractions in the original Yasso20 are not imposed, because they would give Yasso20 fewer degrees of freedom than Yasso07/15 and make the structural intercomparison uninterpretable.
10. **Stick-breaking parameterisation with budget $B = 1 - p_H$.** Guarantees the per-column mass-budget constraint exactly; the remainder of each pool’s outflow is respired as CO_2 .
11. **Weakly informative priors on transfer fractions** ($\sigma_{\text{ppm}} = 1.0$ in unconstrained space). Finnish data drives pool-flow structure, which is the primary structural-comparison axis.
12. **Informative priors on climate and size parameters** (Yasso15/20 from FMI posterior SDs; Yasso07 from literature CIs). Prevents the $\gamma, \sigma_{\text{input}}$ compensation loop and the δ_2 NaN pathology.
13. **Three SOC observation years (1985, 2006, 2024).** The Komeetta campaign (2024) provides a third calibration target. The 1985–2006 interval constrains the accumulation dynamics; the 2006–2024 near-stability constrains long-term decomposition rates and the saturation behaviour. Full Komeetta data description in the data documentation.
14. **SP1, TP2, and TP3 share the Yasso07 ξ formula.** Climate response comparison across the pool-complexity axis is not confounded by approximation-scheme differences.
15. **ICBM topology for TP2.** All litter enters *A*; *H* receives only transferred mass. Eliminates a weakly-constrained input-partitioning parameter and is standard in Nordic forestry SOC models.
16. **Sequential cascade for TP3.** No back-transfers between pools; $A \rightarrow S \rightarrow H$ only. This is the minimal three-pool generalisation of TP2, extending the complexity ladder by one step before the full AWEN topology of Yasso07.

17. **Global σ_{input} .** A single multiplicative litter scaling per model run. Per-species or per-region stratification is a possible extension but is not used in the current pipeline.
18. **No physical-ceiling rejection** in the likelihood. Implausible predictions are penalised via the log-likelihood, not a hard cap. Consistent with Viskari (2022).

14 Software

Component	Details
R	≥ 4.3
BayesianTools	Hartig et al. (2023); DEzs sampler
coda	Gelman–Rubin $\hat{R}$, ESS
compiler	<code>cmpfun()</code> byte-compilation of likelihood
parallel	<code>mclapply()</code> per-plot parallelism
ranger	Random-forest residual analysis
Fortran 90 cores	R CMD SHLIB; <code>.Fortran()</code> interface (Yasso family)
SP1, TP2, TP3	Pure R; no Fortran

Random seed: `set.seed(2025)` before chain initialisation, consistent across all models. Test plot subsampling (when `N_PLOTS_TEST` is set): `set.seed(42)`.

15 References

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